

DANG TRAN NAM

Systems & Network Engineer
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Currently a Software Engineer with a deep passion for the lower levels of the stack. I am actively expanding my expertise into network infrastructure (CCNA) and Linux systems administration (LPI). I specialize in building resilient, high-performance architectures, focusing on OS-level tuning, bare-metal optimization, and robust distributed deployments.

EXPERIENCE

An Binh Hospital

Jan 2026 - Present

Software Engineer Intern

IT ASSET MANAGEMENT (ITAM) – DISTRIBUTED TELEMETRY SYSTEM

- Daemon Architecture & OS APIs:** Developed a lightweight, cross-platform daemon in Go to harvest system telemetry (CPU/RAM/Disk) directly via Windows/Linux OS APIs.
- Network Resilience:** Engineered a Circuit Breaker pattern with exponential backoff and randomized "Jitter" to prevent network thundering herds when thousands of edge agents poll simultaneously.
- System Administration & Deployment:** Configured Windows GPO policies to deploy the agent as a persistent, deep-sleep background service with a strict memory footprint (<15MB RAM).
- Security:** Secured Agent-Server command execution (remote kill switches) using Bearer Tokens and SHA-256 HMAC signature verification.

IT HELPDESK & NETWORK SUPPORT

- Technical Support:** Provided Tier 1 & Tier 2 troubleshooting for hospital staff, resolving hardware, software, and peripheral issues to ensure zero downtime in critical medical environments.
- Network Troubleshooting:** Assisted in deploying and diagnosing local area networks (LAN), verifying switch connectivity, and maintaining stable intranet access across hospital wards.
- Systems Provisioning:** Deployed Windows workstations, managed Active Directory user credentials, and enforced strict hospital data security compliance protocols.

PROJECTS

DONALD VIBE – HIGH-CONCURRENCY INVENTORY ENGINE

Solo Architect

A high-concurrency transactional engine architected to survive "thundering herd" traffic spikes without external cache dependencies.

- OS-Level I/O Tuning:** Configured SQLite in WAL (Write-Ahead Logging) mode and executed C-level PRAGMA tuning to reduce fsync () syscalls, sustaining 10,000+ writes/sec with deterministic sub-0.1ms p95 latency.
- Containerization & Infrastructure:** Engineered ultra-lightweight (~15MB) multi-stage Docker builds on Alpine Linux. Orchestrated multi-tier topologies (Nginx Load Balancer, App Replicas) via Docker Compose.
- Cloud Automation:** Automated AWS service provisioning (S3, SQS) using LocalStack and Bash scripting to ensure strict environment parity across CI pipelines.

SKILLS

Hard Skills	Linux (Arch/Ubuntu), Golang, Rust, JavaScript, Docker, C/C++, Python, AWS, Google Cloud Platform
AI Tooling	GPT, Claude, Gemini Pro, Grok (Prompting, Automation, & API Integration)

EDUCATION

CERTIFICATIONS

- TOEIC (760 R/L)
- AWS Cloud (In Progress)
- Software Engineer Intern Certificate - HackerRank